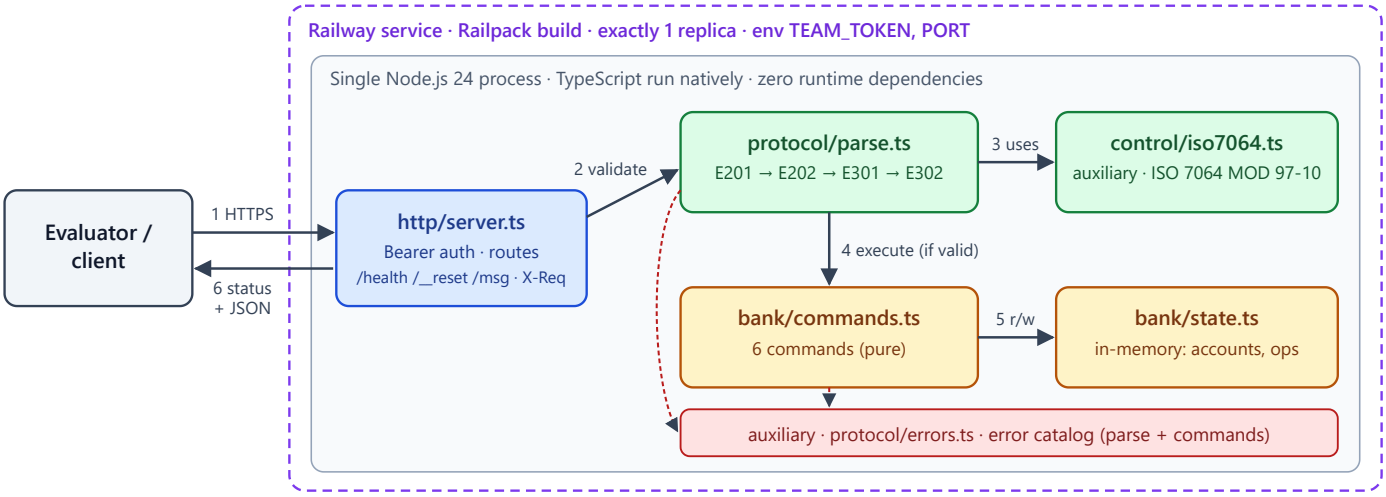
System Design — Reto 2 "La caja negra" replica

Drop-in replacement for an undocumented service: same HTTP status and body for every request, errors included. Node.js 24 + TypeScript, `node:http`, zero runtime dependencies, one in-memory state, deployed as a single Railway instance.



Flow: 1 request → 2 validate message → 3 check control value → 4 execute command → 5 read/update state → 6 response. Labels: observed backed by recorded responses (`pruebas/ejemplos.jsonl`, `experimentos.jsonl`); assumption untested default, isolated in code as `// ASSUMPTION:`.

Components and responsibilities

- `http/server.ts` — only I/O adapter: Bearer auth, routes, JSON body, `X-Req` counter.
- `protocol/parse.ts` — splits `m` on `;` (last field = control); validates format, control value, command (case-insensitive), arity.
- `control/iso7064.ts` — pure: uppercase payload, digits kept, A=1…Z=26, separators dropped; `98 - (N·100 mod 97)`.
- `bank/commands.ts` — PING, ALTA, CONSULTA, DEPOSITO, TRANSFER, ANULA; validates all inputs before mutating.
- `bank/state.ts` — accounts, operations, `AC-NNNN` / `OP-NNNN` counters, reset.
- `protocol/errors.ts` — single catalog of byte-exact bodies (incl. `"recieved"`, trailing space in `"comando desconocido "`).

State and persistence

- In memory only, one state bound to `TEAM_TOKEN`; `POST /__reset` clears it → `200 {"ok":true}` observed.
- Sequential ids from 0001; rejected requests and ANULA consume no id observed.
- ANULA of a TRANSFER debits the destination observed; credits the origin assumption.
- No persistence needed: every evaluation test starts with `POST /__reset` (README).

Authentication, validation, errors

- Wrong token → `401 E001 "no autorizado"` observed; missing token, same assumption. `GET /health` public → 200.
- Control: non-digits → `400 E201`; wrong value → `400 E202` observed, ISO formula confirmed on edge residues.
- Unknown command → `400 E301`; wrong arity → `400 E302`; unknown account → `404 E401`; limit → `422 E506`; funds → `422 E507` observed.
- Order E201 → E202 → E301 → E302, transfer limit `> 100000`, codes E101/E303/E402/E404/E500 assumption.

Technical decisions

- Native TypeScript stripping: no build step, fast start.
- Pure `control` / `bank` + one HTTP adapter: logic testable without a server.
- Golden tests replay the 20 recorded examples and the ISO experiment (`npm test`: 8/8 pass); `tsc --noEmit` clean.
- No LLM inside the replica: behavior is deterministic. Never calls the original.
- Engineering contract in `AGENTS.md` (strict types, no secrets, Definition of Done).

Deployment — single instance

- Railway, Railpack builder (no Docker): `node src/main.ts`, health check `/health`, restart `ALWAYS`, Node `24.x`.
- Exactly 1 replica, no autoscaling, no sleeping: state is per process and each step has 5 s.
- `TEAM_TOKEN` as a service variable, never committed or logged.

Risks and limitations

- Untested error order and invented codes may fail multi-error or edge-case tests.
- The original may change during the day; parity is checked against the 20:00 version.
- A restart loses state mid-sequence; mitigated by one always-on instance.